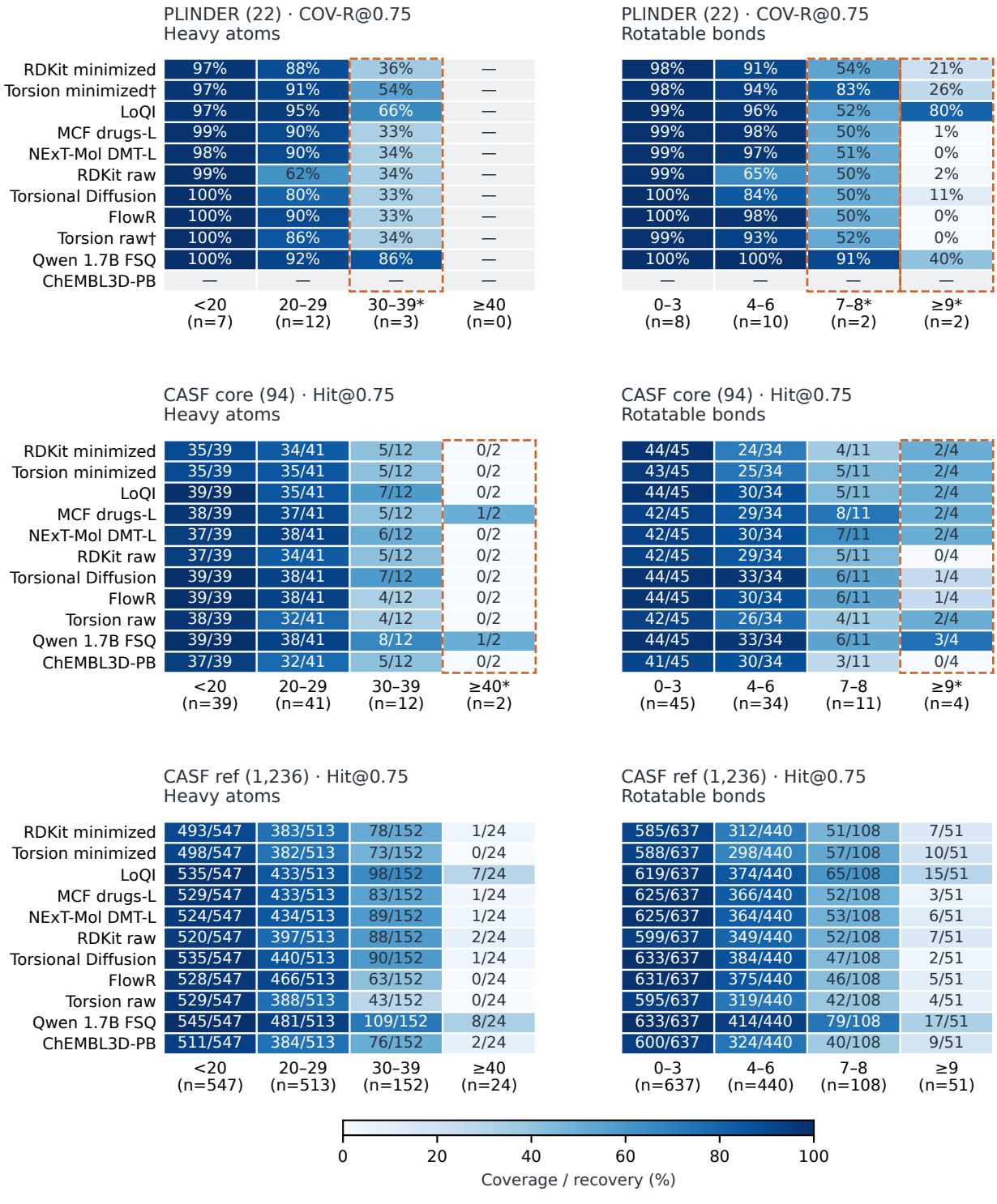
PLINDER (22) · COV-R@0.75
Heavy atoms
RDKit minimized 97% 88% 36% —
Torsion minimized† 97% 91% 54% —
LoQI 97% 95% 66% —
MCF drugs-L 99% 90% 33% —
NExT-Mol DMT-L 98% 90% 34% —
RDKit raw 99% 62% 34% —
Torsional Diffusion 100% 80% 33% —
FlowR 100% 90% 33% —
Torsion raw† 100% 86% 34% —
Qwen 1.7B FSQ 100% 92% 86% —
ChEMBL3D-PB — — — —
<20 (n=7)
20–29 (n=12)
30–39* (n=3)
≥40 (n=0)
PLINDER (22) · COV-R@0.75
Rotatable bonds
98% 91% 54% 21%
98% 94% 83% 26%
99% 96% 52% 80%
99% 98% 50% 1%
99% 97% 51% 0%
99% 65% 50% 2%
100% 84% 50% 11%
100% 98% 50% 0%
99% 93% 52% 0%
100% 100% 91% 40%
— — — —
0–3 (n=8)
4–6 (n=10)
7–8* (n=2)
≥9* (n=2)
CASF core (94) · Hit@0.75
Heavy atoms
RDKit minimized 35/39 34/41 5/12 0/2
Torsion minimized 35/39 35/41 5/12 0/2
LoQI 39/39 35/41 7/12 0/2
MCF drugs-L 38/39 37/41 5/12 1/2
NExT-Mol DMT-L 37/39 38/41 6/12 0/2
RDKit raw 37/39 34/41 5/12 0/2
Torsional Diffusion 39/39 38/41 7/12 0/2
FlowR 39/39 38/41 4/12 0/2
Torsion raw 38/39 32/41 4/12 0/2
Qwen 1.7B FSQ 39/39 38/41 8/12 1/2
ChEMBL3D-PB 37/39 32/41 5/12 0/2
<20 (n=39)
20–29 (n=41)
30–39 (n=12)
≥40* (n=2)
CASF core (94) · Hit@0.75
Rotatable bonds
44/45 24/34 4/11 2/4
43/45 25/34 5/11 2/4
44/45 30/34 5/11 2/4
42/45 29/34 8/11 2/4
42/45 30/34 7/11 2/4
42/45 29/34 5/11 0/4
44/45 33/34 6/11 1/4
44/45 30/34 6/11 1/4
42/45 26/34 4/11 2/4
44/45 33/34 6/11 3/4
41/45 30/34 3/11 0/4
0–3 (n=45)
4–6 (n=34)
7–8 (n=11)
≥9* (n=4)
CASF ref (1,236) · Hit@0.75
Heavy atoms
RDKit minimized 493/547 383/513 78/152 1/24
Torsion minimized 498/547 382/513 73/152 0/24
LoQI 535/547 433/513 98/152 7/24
MCF drugs-L 529/547 433/513 83/152 1/24
NExT-Mol DMT-L 524/547 434/513 89/152 1/24
RDKit raw 520/547 397/513 88/152 2/24
Torsional Diffusion 535/547 440/513 90/152 1/24
FlowR 528/547 466/513 63/152 0/24
Torsion raw 529/547 388/513 43/152 0/24
Qwen 1.7B FSQ 545/547 481/513 109/152 8/24
ChEMBL3D-PB 511/547 384/513 76/152 2/24
<20 (n=547)
20–29 (n=513)
30–39 (n=152)
≥40 (n=24)
CASF ref (1,236) · Hit@0.75
Rotatable bonds
585/637 312/440 51/108 7/51
588/637 298/440 57/108 10/51
619/637 374/440 65/108 15/51
625/637 366/440 52/108 3/51
625/637 364/440 53/108 6/51
599/637 349/440 52/108 7/51
633/637 384/440 47/108 2/51
631/637 375/440 46/108 5/51
595/637 319/440 42/108 4/51
633/637 414/440 79/108 17/51
600/637 324/440 40/108 9/51
0–3 (n=637)
4–6 (n=440)
7–8 (n=108)
≥9 (n=51)
0
20
40
60
80
100
Coverage / recovery (%)